Analyzing Military Health System Pharmacy Technician Attrition Through the Porter–Lawler Satisfaction–Performance Model

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Jenny was a smart and ambitious pharmacy technician who wanted opportunities that could enrich her career at the military treatment facility (MTF). She asked her leadership to attend an online certification program in pharmacy supply chain management through a detailed proposal. Unfortunately, her position description (PD) didn't recognize nor provide compensation for those expanded responsibilities. Additionally, there were no open positions to use those sought after skills. Within weeks of that conversation, Jenny resigned. Her story is about nearly one in four pharmacy technicians nationwide who are leaving their jobs as evidenced by a low career growth satisfaction of 25%, directly threatening the operational readiness of MTFs in the Military Health System (MHS) (American Society of Health-System Pharmacists, 2022; Pharmacy Technician Certification Board, 2022). The purpose of this paper is to examine the root causes of pharmacy technician turnover in MTFs through the Porter–Lawler Satisfaction–Performance Model (PLSPM) by exploring how civilian health systems have engineered successful technician career ladders and then outline practical managerial solutions to maximize technician retention and performance in the MHS.

Discussion and Problem

Over the last two decades, health-system pharmacy has aggressively expanded the role of pharmacists into advanced clinical practice. However, this created an operational paradox: pharmacists can only practice at the top of their license if highly skilled pharmacy technicians absorb the complex responsibilities of the pharmacy. Unfortunately, this created a disparity where 42% of pharmacy technicians cite inadequate pay/compensation as their number one reason for leaving their profession, with another 25% demanding a formal career ladder with

clear promotion paths (American Society of Health-System Pharmacists, 2026). Furthermore, the federal sector has reported that the pharmacy technician occupation series (GS-0661) has been designated as a severe occupational staffing shortage for multiple consecutive years (U.S. Department of Veterans Affairs, 2024, 2025).

Adding to operational frustration is the fragmented reality of regulation and credentialing. While the Office of Personnel Management (OPM) mandates at least one year of specialized pharmaceutical experience to qualify for a GS-04 position (U.S. Office of Personnel Management, 2024), national entry requirements for pharmacy technicians remain a state-by-state patchwork. Professional organizations like the Pharmacy Technician Certification Board (PTCB) have aggressively expanded advanced credentials, such as the Advanced Certified Pharmacy Technician (CPhT-Adv) to foster an enduring professional identify within the technician corps (Pharmacy Technician Certification Board, 2025). Currently, the direct consequence of an operational system that demands advanced technical performance while offering no structural path for professional growth is less than 40% of technicians remain with their current employer for greater than six years(Pharmacy Technician Certification Board, 2025).

Theoretical Framework and Application

Porter and Lawler: Satisfaction–Performance Model (1967)

In 1967, organizational researchers Lyman Porter and Edward Lawler III combined elements of the Expectancy Theory and Equity Theory into a new model that considered the relationship between job performance and employee satisfaction [figure 1] (Meese & Borkowski, 2026). Their research found that roles and skills were important factors for high performance, and if they believed the reward was valuable and tangible. They also found that employee

satisfaction is derived from the perceived fairness of the intrinsic and extrinsic rewards from their overall performance.

Empirical Limits of the Performance–Satisfaction Relationship in Healthcare

Porter and Lawler (1967) found that intrinsic rewards had the highest correlation with both job satisfaction and performance. This was further validated by a comprehensive qualitative and quantitative review which found a strong correlation ($r = 0.30$) to support this relationship (Judge et al., 2001). This means that the inputs of job performance (ie abilities and traits, and role perceptions) factor more strongly into the intrinsic satisfaction that good job performance sustains, despite extrinsic factors. While Lawler and Porter's Satisfaction–Performance Model has not been directly tested in pharmacy technicians in the MTF there are a few studies that have generalizability to this population.

In the same review, Judge et al. also noted highly complex roles, such as nursing, showed a higher performance-satisfaction relationship of ($r = 0.37$) suggesting that in high-demand patient care environments, clinical performance is driven by professional duty, protocols, and role expectations rather than personal job satisfaction. In another study by Mowday, Steers, and Porter (1979), they found that hospital workers and psychiatric technicians had comparatively low organizational commitment that was less responsive to pay and performance that was independent of commitment. What all of this means is that we must look at different approaches to motivate pharmacy technicians than extrinsic rewards. In Lawler and Porter's paper (1967), they suggest that good performance is rewarding for itself, shifting the conservation away from perceived equitable rewards to upstream performance factors such as 'abilities and traits' and 'role perceptions'. This means that healthcare leaders must look beyond meager bonuses (1-2%)

and develop stimulating and progressive career development, that foster this intrinsic reward rather than unachievable financial incentives (Perry et al., 2009).

Federal Employment does Not Support Instrumentality and Perceived Equitable Rewards

Lawler and Porter's model abruptly diminishes when applied to the General Schedule (GS) system of the U.S. Government. Here, promotions and extrinsic rewards are directly tied to time in service and adequate performance. Inconsistent financial appropriations and rigid standardization do not leave space for consistent rewards or autonomy - both noted as key factors of commitment and satisfaction (Judge et al., 2001; Mowday et al., 1979). In a review of 57 reviews of federal pay programs, Perry et al. (2009) further established that pay-for-performance structure showed there was no demonstratable difference in performance, as merit payouts were negligible and subjective appraisals eroded trust and intrinsic motivation. They concluded that the 'payroll cost-containment' strategies of federal employment will likely never reach the reported 10-15% pay expectancy to motivate employees, requiring future incentives to rely on improving performance through clear performance standards, objective goal setting, and job design rather than financial incentive schemes.

Strategic Managerial Solutions

Now given that the instrumentality of extrinsic rewards does not work within federal civil service constraints, leaders must identify performance modifiers that are practical and value-driven investments in both pharmacy staff and the Military Health System. In similar international public healthcare systems, evidence shows that pay and promotion opportunities contribute little to performance and satisfaction under rigid compensation systems (Al-Ahmadi, 2009; McDermott et al., 2025). Instead, National Health System (NHS, UK) technicians reported greater satisfaction from coworker cohesion (80%) and task autonomy (72%), reinforcing the

importance of intrinsic and relational rewards (McDermott et al., 2025). These findings directly validate what Frazier et al. (2022) observed across Military Treatment Facilities where MTF pharmacies achieving the highest patient satisfaction scores demonstrated success through proactive leadership behaviors: cultivating staff morale, maintaining transparent communication, enforcing role clarity, and providing visible leadership recognition (Frazier et al., 2022). In the PLSPM framework, these managerial actions directly shape two vital upstream inputs – role perceptions and abilities and traits – proving that leadership engagement can build high performance even when extrinsic compensation remains fixed.

Role Perceptions

Within the Porter–Lawler framework, role perceptions determine whether an employee understands how to channel their effort into effective, rewarding performance. Several progressive civilian healthcare systems overcome this flat career architecture by developing internal training programs, a multi-stage clinical career ladder [Figure 2], and defining progressive intra- and inter-role competencies [Figure 3] which cut technician turnover by over 50% (down to 13.0-17.1%) and surged employee satisfaction to 84% (O'Neil et al., 2023; Thames et al., 2024). Implementing similar career models within the MHS would not require an act of Congress, as the DHA has full authorization to modernize its core PDs by rewriting the standard roles to accommodate progressive advancement, an entry level GS-04 position can grow and specialize into sterile compounding or clinical support roles (GS-06/07) or Ambulatory technician level one, two, or three with demonstrated skills and abilities.

Skills and Abilities

Under the statutory constraints of federal healthcare, one way to formally recognize legitimate credentials and certifications is to leverage external experts, such as the PTCB, who

can certify and credential pharmacy technicians, as implemented in the joint Military Occupation Specialty (MOS) pharmacy specialist series. This could mirror the board certification pathways through the Army's Health Professions Special and Incentive (HPSI) Pay program, where military pharmacists and other healthcare specialties can receive specialized bonuses or pay incentives when they hold advanced board certifications (U.S. Department of Defense, 2025). Another tangible military example of expanding technician role scope to enhance operational efficiency is the Tech-Check-Tech (TCT) practice model. Under this model, pharmacy leaders discovered that certified technicians could independently verify nearly 50% of daily refill volume (Lerdahl et al., 2023). By expanding technician autonomy and clinical responsibility, TCT strengthens role perceptions and abilities, translating effort into measurable performance and the intrinsic rewards that sustain organizational commitment.

Recommendations

While enterprise-wide staffing deficits remain beyond the scope of this paper, workforce shortages compound pharmacy technician stress, accelerating burnout and a severe decoupling of effort from perceived equitable rewards. However, there are tangible ways that federal healthcare leaders can target the PLSP mechanisms to improve performance and satisfaction within the MHS. Consistent with workforce priorities outlined in the Pharmacy Technician Certification Board and the American Society of Health-System Pharmacists' workforce development standards, DHA and MTF leadership must consider the following recommendations [see Appendix A: Figure 4 and Appendix B for further analysis] (ASHP, 2026; PTCB, 2025):

#	Strategic Recommendation	Responsible Levels/Leaders	Mechanism / Levers	PLSPM Theoretical Impact
1	**Baseline Certification Funding**	MTF Pharmacy Leadership	Protected Admin Time; Tuition/Exam Reimbursement	**Abilities & Traits** (Elevates foundational competence)
2	**Targeted Retention Incentives**	MTF Executive / Resource Mgmt	5 CFR § 575; Special Act & Performance Awards	**Perceived Equitable Rewards** (Re-establishes instrumentality)
3	**Enterprise JKO Modules**	DHA Pharmacy Operations Division	JKO LMS; JMESI-style credentials; PTCB/ASHP	**Abilities & Traits** (Removes cost & access barriers)
4	**Standardized Career Ladders**	DHA Manpower & MTF Leadership	Standardized PD Revisions; DPMAP Annual Appraisals	**Role Perceptions** (Clarifies promotion trajectories)

Conclusion

Pharmacy technicians cannot sustain high clinical workloads while bound to rigid General Schedule pay bands and flat career architectures. As demonstrated through the Porter–Lawler framework, demanding advanced technical performance without viable paths for professional advancement severs the critical link between employee effort, perceived equity, and job satisfaction. Although statutory constraints preclude reforming federal pay systems, the Defense Health Agency has the immediate authority to reshape technician role perceptions and cultivate abilities through modernized position descriptions, enterprise-hosted credentials on JKO, and targeted retention incentives. Investing in progressive career development transforms passive dispensers into empowered clinical teammates, driving down costly attrition and safeguarding pharmacy operations. Ultimately, prioritizing these managerial pathways changes more than retention statistics, it reshapes careers. In an operational system that actively validates advanced competencies and provides clear pathways for growth, dedicated technicians like Jenny do not resign, staffing stabilizes, and military medical readiness is sustained at the highest levels.

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Supplemental Material

Appendix A: Figures and Graphics

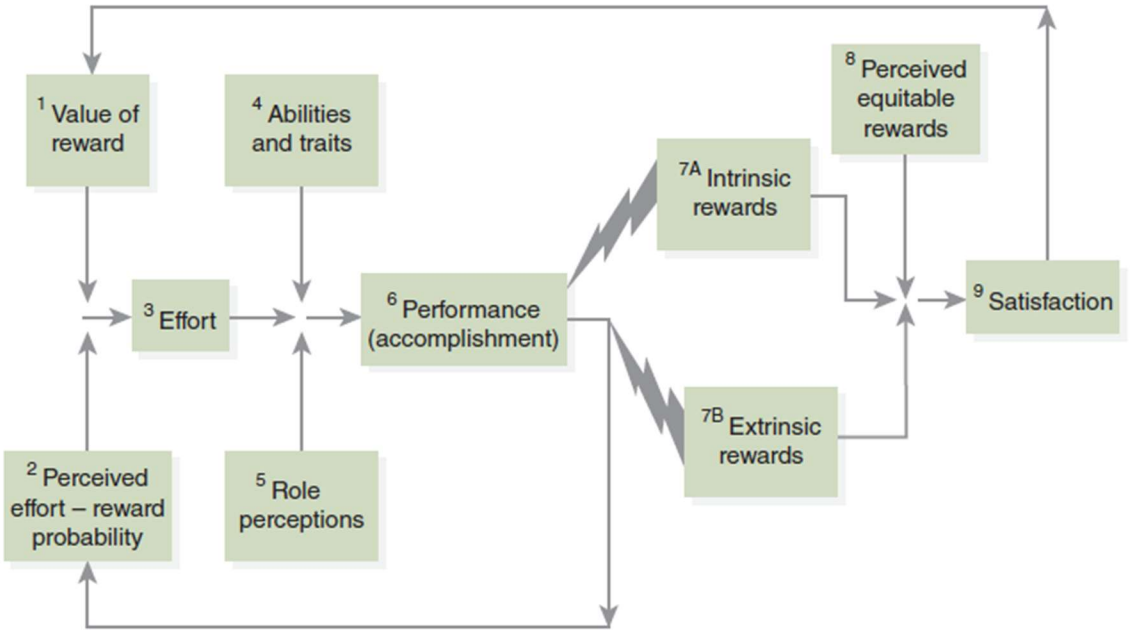


Figure 1. Porter-Lawler Satisfaction-Performance Model. (Meese & Borkowski, 2026)

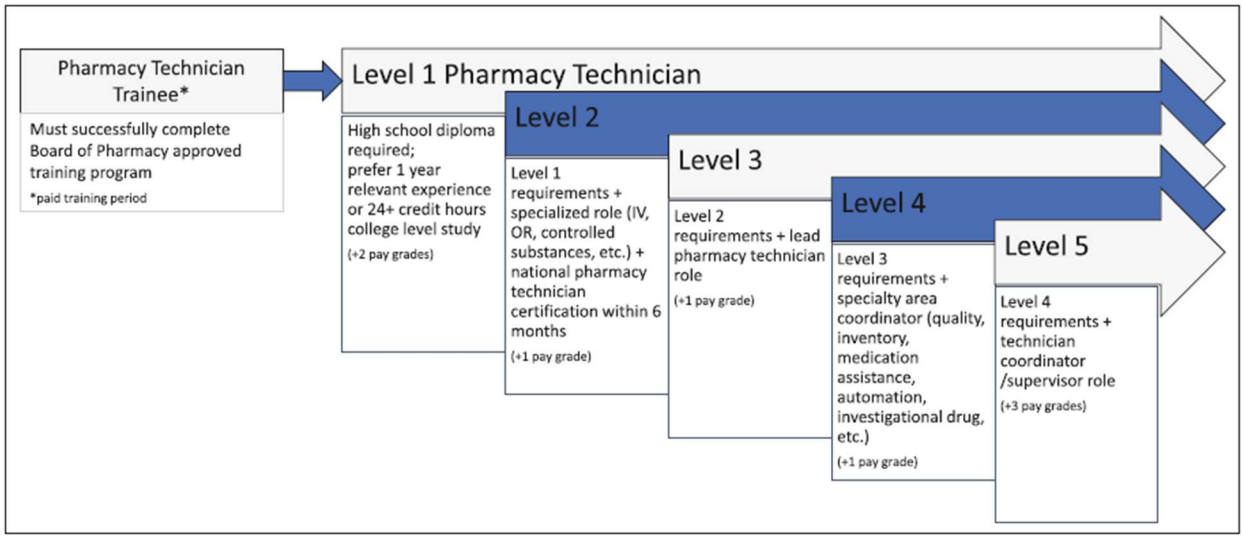


Figure 2. UF Health Shands Hospital pharmacy technician career ladder (Thames et al., 2024)

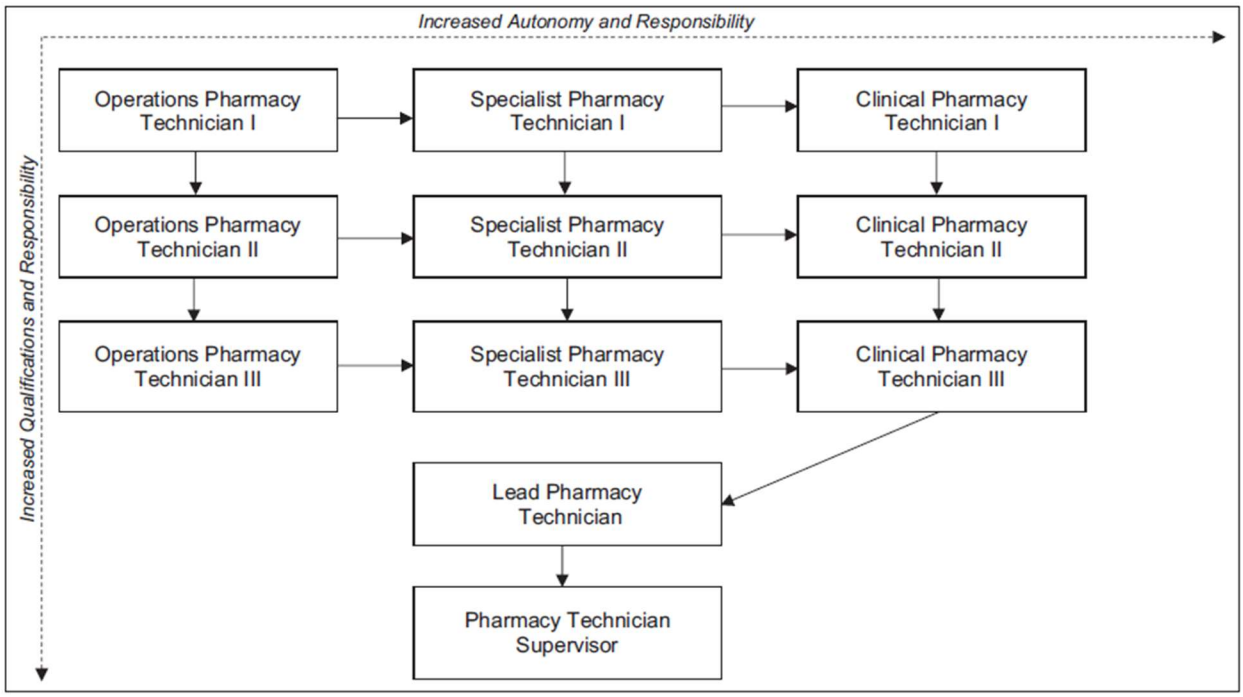
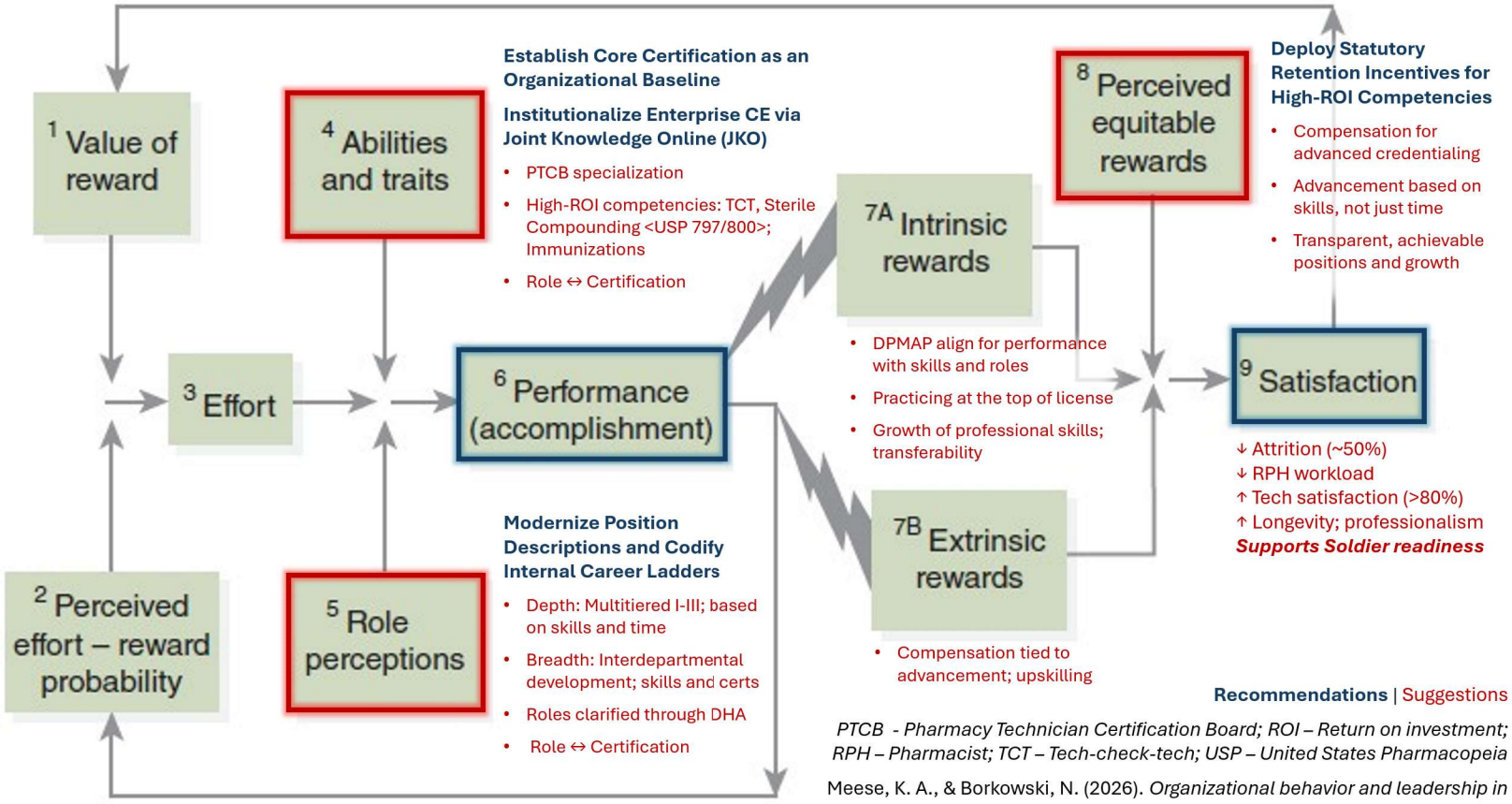


Figure 3. Pharmacy technician career structure (postimplementation) (O'Neil et al., 2023)

Figure 4. Porter-Lawler Satisfaction-Performance Model for Pharmacy Technicians in the Military Health System (VanBemmel, 2026)

(next page)

Porter-Lawler Satisfaction-Performance Model for Pharmacy Technicians in the Military Health System



PTCB - Pharmacy Technician Certification Board; ROI – Return on investment; RPH – Pharmacist; TCT – Tech-check-tech; USP – United States Pharmacopeia

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Appendix B: Recommendations

Consistent with workforce priorities outlined in the Pharmacy Technician Certification Board (PTCB, 2025) and the American Society of Health-System Pharmacists' workforce development standards (ASHP, 2026), DHA and MTF leadership must consider the following recommendations:

1. **Establish Core Certification as an Organizational Baseline**: Establish dedicated, protected administrative time and educational funding for GS pharmacy technicians to attain and initial CPhT credentials, treating national board certification not as a personal civilian expense, but as a vital operational baseline for MHS patient safety (ASHP, 2026; PTCB, 2025).
2. **Deploy Statutory Retention Incentives for High-ROI Competencies**: MTF leadership should utilize existing federal flexibilities – specifically retention incentives and performance awards under 5 CFR § 575 – to reward advanced clinical competencies with documented institutional return on investment. Targeted milestones should include Tech-Check-Tech (TCT/TPV) validation, USP <797>/<800> hazardous and sterile compounding (CSPT), vaccine administration, and CPhT-Adv status. Linking these competencies to formal, recurring retention allowances recognizes advanced technical mastery and closes the instrumentality gap without requiring an unfeasible OPM reclassification.
3. **Institutionalize Enterprise CE via Joint Knowledge Online (JKO):** The Defense Health Agency should eliminate individual financial barriers to recertification by hosting accredited continuing education directly on enterprise platforms like JKO. Modeled after the Joint Medical Executive Skills Program (JMESI), DHA can partner with PTCB and

ASHP to curate modular micro-credentials. Enterprise LMS hosting centralizes compliance tracking, provides verified training jackets for supervisors, and eliminates out-of-pocket costs for frontline civilian personnel.

4. **Modernize Position Descriptions and Codify Internal Career Ladders:** In step with the national shift toward employer-based training (accounting for 33% of technician education nationally; PTCB, 2025), DHA and local MTF departments must rewrite static GS-0661 Position Descriptions to establish structured intra-role tiers (e.g., Ambulatory Tech I/II, Sterile Compounding Specialist, Lead Vault Technician). Formally integrating these tier milestones into annual Defense Performance Management and Appraisal Program (DPMAP) performance plans creates clear role perceptions and establishes transparent criteria for institutional advancement.

Appendix C: Acknowledgments

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